A degree-two neighborhood bound for total domination in trees

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Abstract

Let T be a finite simple tree of order at least three, let $D_2(T)$ be the set of its degree-two vertices, and put $b(T) = |N_T(D_2(T)) \setminus D_2(T)|$, where N_T denotes the open neighborhood. We prove that every total dominating set D of T satisfies $3|D| \geq 2b(T) + 2$. Consequently, $\gamma_t(T) \geq \lceil (2b(T)+2)/3 \rceil$, which implies the inequality recorded as Conjecture 348 in *Written on the Wall II*. The proof is a structural induction using seven explicit bounds for rooted subtrees, with no upper bound on the order or degrees. The additive constant is best possible with the coefficients 3 and 2 fixed, as witnessed by P_3 and P_4. An explicit family of connected graphs shows that the tree hypothesis is essential even for the original inequality. A supplementary standard-library Python package checks all labelled trees of orders three through seven and includes a second algorithm for exact total domination. These finite checks are separate from the proof of the universal statement.

Keywords. Total domination; trees; degree-two vertices; Graffiti.pc; rooted-tree induction.
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1 Introduction and statement

Graffiti.pc generated a number of conjectures concerning the total domination number of a graph. DeLaViña's collection *Written on the Wall II* [1] records the following assertion as number 348, dated 18 February 2009: if T is a tree on $n > 2$ vertices, then

$$\gamma_t(T) \geq \frac{2}{3} |N_T(D_2(T)) \setminus D_2(T)|, \qquad D_2(T) = \{v \in V(T) : \deg_T(v) = 2\}. \tag{1}$$

The original entry uses γ_T for the total domination number; we use the customary notation γ_t throughout. The neighborhood and set subtraction in (1) are those of the original definitions. In particular, degree-two vertices lying in $N_T(D_2(T))$ do not contribute to the quantity on the right.

The study of Graffiti.pc bounds for total domination in trees includes the paper of DeLaViña, Larson, Pepper, and Waller [2]. Desormeaux and Henning [3] also investigated Graffiti.pc lower bounds for total domination. We supply here a direct proof of (1) with an additional constant. No result from either paper is used as a premise. The source identification and the limits of the literature comparison are stated in Appendix C; no assertion of first priority is made.

All graphs in this paper are finite, simple, and undirected. A tree is a connected graph containing no cycle. Its order is $|V(T)|$. For a graph G and a vertex v, let

$$N_G(v) = \{w \in V(G) : vw \in E(G)\}, \qquad \deg_G(v) = |N_G(v)|.$$

For $X \subseteq V(G)$, define its open neighborhood by

$$N_G(X) = \bigcup_{x \in X} N_G(x).$$

This union can intersect X; a vertex does not belong to the union merely because it belongs to X. A set $D \subseteq V(G)$ is *total dominating* if $N_G(v) \cap D \neq \varnothing$ for every $v \in V(G)$. Thus every vertex of D itself must have a neighbor in D. The total domination number $\gamma_t(G)$ is the minimum size of such a set. It exists for any graph with no isolated vertex, since its full vertex set is total dominating. A tree of order at least two has no isolated vertex.

For a tree T, set

$$D_2(T) = \{v \in V(T) : \deg_T(v) = 2\}, \qquad B(T) = N_T(D_2(T)) \setminus D_2(T), \qquad b(T) = |B(T)|. \tag{2}$$

Equivalently, $v \in B(T)$ if and only if $\deg_T(v) \neq 2$ and v has a neighbor of degree two. This follows directly from the open neighborhood and the set difference.

Theorem 1.1. *Let T be a tree of order at least three. Every total dominating set D of T satisfies*

$$3|D| \geq 2b(T) + 2. \tag{3}$$

Corollary 1.2. *Every tree T of order at least three satisfies*

$$\gamma_t(T) \geq \left\lceil \frac{2b(T) + 2}{3} \right\rceil. \tag{4}$$

In particular, (1) holds on its full stated domain.

Proof. Apply Theorem 1.1 to a minimum total dominating set. Division by three gives $\gamma_t(T) \geq (2b(T) + 2)/3$. Since $\gamma_t(T)$ is an integer, this implies (4). Finally, $(2b(T) + 2)/3 \geq 2b(T)/3$, proving (1). □

2 The rooted-subtree bounds

Fix a tree T of order at least three, a total dominating set D, and initially an arbitrary root r. Between any two vertices of a tree there is a unique simple path: connectedness gives a path, and two different simple paths would contain a cycle. For $v \neq r$, its *parent* $\mathrm{par}(v)$ is the neighbor on its path to r. The other neighbors are its *children*, whose set is $\mathrm{Ch}(v)$. A nonroot vertex of degree d has exactly $d - 1$ children. Let T_v be the subtree induced by v and its descendants, including v. The sets $V(T_w)$ for $w \in \mathrm{Ch}(v)$ are pairwise disjoint and partition $V(T_v) \setminus \{v\}$, because each descendant path to v passes through a unique first child.

For every vertex x, put $s_x = \mathbf{1}_{\{x \in D\}}$. For $v \neq r$, put

$$p_v = s_{\mathrm{par}(v)}, \qquad q_v = \mathbf{1}_{\{\deg_T(\mathrm{par}(v))=2\}}.$$

Define the subtree weight

$$F(v) = \sum_{x \in V(T_v)} \left(3s_x - 2\mathbf{1}_{\{x \in B(T)\}}\right). \tag{5}$$

All degrees, and all membership tests for $B(T)$, refer to the original tree T. They are never recomputed in T_v. The parent of v is outside the sum in (5). The disjoint partition above gives the exact identity

$$F(v) = 3s_v - 2\mathbf{1}_{\{v \in B(T)\}} + \sum_{w \in \mathrm{Ch}(v)} F(w). \tag{6}$$

Two consequences of total domination will be used repeatedly. If $v \neq r$ and $p_v = 0$, then v has at least one selected child: its neighbors are its parent and children, and the parent is not in D. This holds whether or not $s_v = 1$. A nonroot leaf must have $p_v = 1$, since its parent is its only neighbor.

$\deg_T(v)$	s_v	Parent condition	Bound for $F(v)$
1	0 or 1	$p_v = 1$	$3s_v - 2q_v$ (exact)
2	0	Any	2
2	1	$p_v = 0$	4
2	1	$p_v = 1$	1
≥ 3	0	Any	2
≥ 3	1	$p_v = q_v = 1$	1
≥ 3	1	All other cases	2

Table 1: Lower bounds in the original tree. *Any* allows either value of each unspecified parent indicator, subject to total domination.

Lemma 2.1. *For every nonroot vertex v, the applicable row of Table 1 is a lower bound for $F(v)$. A nonroot leaf with $p_v = 0$ is impossible.*

Proof. We induct on the height of T_v, the maximum number of edges on a path from v to a descendant. A nonroot vertex of height zero has no child and therefore degree one. At positive height, every child has smaller height, so all applicable table rows are already valid for that child. Each case below uses only these previously established child bounds and (6).

Case 1: $\deg_T(v) = 1$. The only neighbor of v is its parent. Hence $p_v = 1$. Since $v \notin D_2(T)$, it belongs to $B(T)$ precisely when its parent has degree two, that is, when $q_v = 1$. It has no child, and (5) gives $F(v) = 3s_v - 2q_v$. This proves the induction base and the first row.

Case 2: $\deg_T(v) = 2$. There is exactly one child w. The subtraction in (2) ensures $v \notin B(T)$. Thus

$$F(v) = 3s_v + F(w), \qquad (p_w, q_w) = (s_v, 1). \tag{7}$$

There are three selection conditions at v.

If $s_v = 0$, then $p_w = 0$. The child w cannot be a leaf: its only neighbor would be the unselected vertex v. For a degree-two child with $p_w = 0$, the induction hypothesis gives $F(w) \geq 2$ if $s_w = 0$, and $F(w) \geq 4$ if $s_w = 1$. For a child of degree at least three, it gives $F(w) \geq 2$ when $s_w = 0$, and also $F(w) \geq 2$ when $s_w = 1$, since $p_w = 0$ excludes the sixth row. These are all possible child degrees. Therefore $F(w) \geq 2$, and (7) gives $F(v) \geq 2$.

If $s_v = 1$ and $p_v = 0$, total domination of v forces $s_w = 1$. Now $(p_w, q_w) = (1, 1)$. A selected leaf has weight $3 - 2 = 1$ by Case 1. A selected degree-two child has weight at least 1 by the fourth row of the induction hypothesis. A selected child of degree at least three has weight at least 1 by its sixth row. Hence $F(w) \geq 1$ and $F(v) = 3 + F(w) \geq 4$.

If $s_v = 1$ and $p_v = 1$, again $(p_w, q_w) = (1, 1)$, but s_w may have either value. The possible child bounds are -2 for an unselected leaf, 1 for a selected leaf, 2 for an unselected degree-two child, 1 for a selected degree-two child, 2 for an unselected child of degree at least three, and 1 for a selected child of degree at least three. All are at least -2. Consequently $F(v) = 3 + F(w) \geq 1$. This proves the three degree-two rows.

Case 3: $\deg_T(v) \geq 3$ *and* $s_v = 0$. The vertex v has at least two children. Each child w has $(p_w, q_w) = (0, 0)$, since v is unselected and not of degree two. No child can be a leaf, by total domination of that child. For a degree-two child, the second and third rows give bounds 2 and 4, according to its selection. For a child of degree at least three, the fifth and seventh rows both give 2; the sixth row cannot apply because $p_w = 0$. Thus every child contributes at least 2. The contribution of v itself is at least -2, and so

$$F(v) \geq -2 + 2|\operatorname{Ch}(v)| \geq -2 + 2 \cdot 2 = 2.$$

There is no upper bound on the number of children.

Case 4: $\deg_T(v) \geq 3$ *and* $s_v = 1$. Each child w has $(p_w, q_w) = (1, 0)$. The induction hypothesis gives this exhaustive list of child contributions:

	$s_w = 0$	$s_w = 1$
$\deg_T(w) = 1$	0	3
$\deg_T(w) = 2$	≥ 2	≥ 1
$\deg_T(w) \geq 3$	≥ 2	≥ 2

Every child weight is nonnegative, every selected child has weight at least 1, and every degree-two child has weight at least 1.

If $q_v = 1$, the parent of v has degree two. Since v does not have degree two, $v \in B(T)$, and its contribution is $3 - 2 = 1$. If $p_v = 1$, the nonnegative child sum gives $F(v) \geq 1$, as in the sixth row. If $p_v = 0$, total domination of v requires a selected child. That child contributes at least 1, so $F(v) \geq 1 + 1 = 2$, as in the seventh row.

If $q_v = 0$, the parent of v does not have degree two. Therefore $v \in B(T)$ if and only if at least one child has degree two, since its parent and children are all its neighbors. When such a child exists, v contributes 1, that child contributes at least 1, and all other contributions are nonnegative. Hence $F(v) \geq 2$. Otherwise $v \notin B(T)$, its contribution is 3, and $F(v) \geq 3 \geq 2$. This proves the seventh row for either value of p_v.

The cases cover every nonroot vertex and every feasible selection of its indicators. The induction is complete. □

3 Closing the induction at a leaf

Proof of Theorem 1.1. A finite tree of order at least two has a leaf. Choose a longest simple path. An endpoint has no neighbor outside the path, since that would extend it, and no neighbor on the path other than the next vertex, since that would create a cycle. Its degree is one.

Choose a leaf r as the root and let v be its unique neighbor. Total domination of r forces $s_v = 1$. Since $|V(T)| \geq 3$, we have $\deg_T(v) \geq 2$: if both endpoints of rv had degree one, connectedness would force $V(T) = \{r, v\}$. Also $p_v = s_r$ and $q_v = 0$, since $\deg_T(r) = 1$. All vertices except r belong to T_v, and therefore

$$3|D| - 2b(T) = 3s_r - 2\mathbf{1}_{\{r \in B(T)\}} + F(v). \tag{8}$$

If $\deg_T(v) = 2$, then $r \in B(T)$, since r is a leaf adjacent to a degree-two vertex. When $s_r = 0$, the third row of Table 1 gives $F(v) \geq 4$, and

$$3|D| - 2b(T) \geq -2 + 4 = 2.$$

When $s_r = 1$, the fourth row gives $F(v) \geq 1$, and

$$3|D| - 2b(T) \geq (3 - 2) + 1 = 2.$$

If $\deg_T(v) \geq 3$, then $r \notin B(T)$, because its only neighbor does not have degree two. Since $s_v = 1$ and $q_v = 0$, the seventh row gives $F(v) \geq 2$ for either value of $p_v = s_r$. Equation (8) yields

$$3|D| - 2b(T) \geq 3s_r + 2 \geq 2.$$

These are all possibilities for v. The argument applies to the arbitrary total dominating set D, proving (3). □

Remark 3.1. The proof does not require D to be minimum, inclusion-minimal, or free of leaves. The seven local bounds apply to the actual selection of any total dominating set; no exchange argument is needed.

4 Equality examples and the tree hypothesis

Let P_m be the path with vertices $z_1, \ldots, z_m$ and edges $z_i z_{i+1}$ for $1 \le i < m$.

Proposition 4.1. *The additive constant 2 in (3) is best possible among uniform bounds with the coefficients 3 and 2 fixed. Equality holds for P_3 and P_4 with a minimum total dominating set.*

Proof. For P_3, the only degree-two vertex is z_2, and $B(P_3) = \{z_1, z_3\}$. The set $\{z_1, z_2\}$ is total dominating: z_1 and z_2 are adjacent to one another, and z_3 is adjacent to z_2. No singleton is total dominating in a simple graph, because its selected vertex has no selected neighbor. Hence $\gamma_t(P_3) = 2$ and $3\gamma_t(P_3) - 2b(P_3) = 6 - 4 = 2$.

For P_4, $D_2(P_4) = \{z_2, z_3\}$ and $N(D_2(P_4)) = V(P_4)$. Subtracting $D_2(P_4)$ gives $B(P_4) = \{z_1, z_4\}$. The set $\{z_2, z_3\}$ is total dominating; the singleton obstruction gives $\gamma_t(P_4) = 2$. Again $3\gamma_t(P_4) - 2b(P_4) = 2$. Either example excludes a uniform additive constant greater than 2. □

Remark 4.2. These examples establish sharpness of the additive constant with the displayed coefficients fixed. They do not establish asymptotic optimality of $2/3$ as $b(T)$ grows. Trees without degree-two vertices have $b(T) = 0$ and are included in the theorem. The tree K_2 is outside the domain of (1), but satisfies (3) as well: $b(K_2) = 0$ and $\gamma_t(K_2) = 2$. The one-vertex tree has no total dominating set.

In the following proposition, $D_2(G)$ and $B(G)$ have the definitions in (2) with G replacing T.

Proposition 4.3. *For every integer $k \ge 3$, there is a connected simple graph G_k of order $3k + 2$ with $\gamma_t(G_k) = 2$ and $|B(G_k)| = k + 1$. Even (1) consequently fails if its tree hypothesis is replaced by connectedness.*

Proof. Take distinct vertices

$$V(G_k) = \{u, v\} \cup \{c_i, x_i, y_i : 1 \le i \le k\}$$

and precisely the edges

$$E(G_k) = \{uv\} \cup \bigcup_{i=1}^{k} \{vc_i, ux_i, uy_i, c_i x_i, c_i y_i\}.$$

All endpoints are distinct within each edge, and no edge is repeated. Each c_i is joined to v, each x_i, y_i to u, and u to v, so the graph is simple and connected. The distinct vertices u, v, c_i, x_i form a cycle of length four.

The degrees are

$$\deg(u) = 2k + 1, \quad \deg(v) = k + 1, \quad \deg(c_i) = 3, \quad \deg(x_i) = \deg(y_i) = 2.$$

For $k \ge 3$, the degree-two vertices are exactly the x_i, y_i. Their neighborhoods are $\{u, c_i\}$; thus $B(G_k) = \{u, c_1, \ldots, c_k\}$ and $|B(G_k)| = k + 1$. The set $\{u, v\}$ is total dominating: the selected vertices dominate each other, each c_i has selected neighbor v, and each x_i, y_i has selected neighbor u. The singleton obstruction shows $\gamma_t(G_k) = 2$. The extended version of (1) would require $6 \ge 2(k + 1)$, which is false for $k \ge 3$. □

5 Supplementary exact verification

The proof of Theorem 1.1 has no computational premise. The accompanying package supplies finite checks of the definitions, local bounds, examples, and minimum values. We specify the checks so that they can be reproduced without an internal workspace or an external service.

5.1 Representation and exhaustive checks

Vertices are labelled $0, \ldots, n-1$. A subset is an integer mask whose bit i is one exactly when vertex i is selected. The neighborhood mask A_i has a one at each neighbor of i. A mask M is accepted as total dominating precisely when

$$A_i \,\&\, M \neq 0 \quad \text{for every } i, \tag{9}$$

where & is bitwise intersection. This is the definition of total domination, including at selected vertices. Degrees are the numbers of one-bits in A_i. If Q is the degree-two mask and U the bitwise union of its vertices' neighborhood masks, then the boundary mask is $U \,\&\, \neg Q$. Since U has no bit outside $0, \ldots, n-1$, Python's unbounded integer complement adds no vertex.

The routine `adjacency` rejects order less than three, invalid endpoint labels, loops, repeated edges, and disconnected inputs. In tree mode it requires exactly $n-1$ edges. This recognizes the tree domain: from one vertex in a connected graph, repeatedly choose an edge to a new vertex. These edges form a connected acyclic spanning subgraph with $n-1$ edges. If the original graph has $n-1$ edges, it equals that subgraph. Conversely, leaf deletion and induction show that every tree of order n has $n-1$ edges.

For each $n = 3, \ldots, 7$, the program decodes every length-$(n-2)$ sequence over $\{0, \ldots, n-1\}$ by the Prüfer procedure. Appendix A proves that this gives every labelled tree exactly once. For each tree, all 2^n masks are tested. Every accepted total dominating set D is checked for $3|D| \geq 2b(T) + 2$. The minimum accepted size is compared with the four-state optimizer of Appendix B. At orders three through five, every accepted set is also checked against every nonroot bound in Table 1, under every possible root, using original-tree degrees and boundary membership.

Order	Labelled trees	Subsets tested	Accepted TD sets	Equality trees
3	3	24	9	3
4	16	256	76	12
5	125	4 000	855	60
6	1 296	82 944	13 626	120
7	16 807	2 151 296	277 431	210
Total	18 247	2 238 520	291 997	405

Table 2: Exhaustive labelled-tree checks. An equality tree satisfies $3\gamma_t(T) = 2b(T)+2$; different labellings are counted separately.

Table 2 gives the frozen expected output. There are 18 066 further local-bound checks under all roots at orders three through five. A further 70 instances consist of a path, a star, and twelve Prüfer trees for each order in $\{8, 12, 20, 50, 100\}$. One generator `random.Random(20261013)` is used, visiting these orders in the displayed order; each code is obtained by $n-2$ calls to `randrange(n)`. These larger cases use exact optimization and check the selected-set witnesses. They constitute a finite sample.

5.2 Calibration and reproducibility

Before the enumeration, calibration tests check the minimum values $2, 2, 3, 4, 4$ for P_3 through P_7 by subset enumeration and the optimizer. They accept an adjacent pair on P_3, reject its middle singleton, and reject $\{z_1, z_3\}$ on P_4. They check the degree-two and boundary masks of P_3 and P_4. On P_6, $|N(D_2)| = 6$ but $|B| = 2$: omitting the subtraction would incorrectly demand $3 \cdot 4 \geq 2 \cdot 6 + 2$.

The local checker is deliberately given a damaged bound function that replaces the unselected-leaf value -2 by -1 when the parent has degree two. For the actual P_3 selection

$\{z_1, z_2\}$, root z_1 makes z_3 an unselected leaf of weight -2; the checker must reject the damaged bound. Invalid graph inputs exercise the domain rejections. The eleven-vertex G_3 is accepted as a connected graph and rejected as a tree. For orders three through five, the Prüfer edge sets are compared with the connected graphs obtained by enumerating every $(n-1)$-element subset of all possible edges; the two sets coincide.

The archive includes the complete LaTeX source, its compiled PDF, the verifier and replay driver, frozen expected results, source metadata, and a file-digest manifest. The computational files are

```
scripts/verify348.py  reproduce.py
evidence/verification348.json  manifest348.json
```

Python 3.10 or later, with optimization disabled, suffices for verification; no third-party Python package or network is used. From the extracted directory, run

```
python -B -X utf8 reproduce.py
```

The driver verifies every file digest listed in the SHA-256 manifest, copies those files to a fresh temporary directory, runs the verifier there, and compares the entire output JSON with the frozen report. A mismatch or failed check gives a nonzero exit status. Success prints `PASS` and the aggregate counts. Assertions must remain enabled; the verifier explicitly refuses optimized execution. The README gives direct verifier and LaTeX build commands.

Subset enumeration, optimization, and direct edge-set enumeration differ in mechanism, but share graph and domination definitions in this implementation. They are not an external independent review or a formal proof-assistant verification. Hashes establish byte identity relative to the supplied manifest, rather than mathematical correctness or priority. The universal warrant is the induction of Lemma 2.1 and the proof of Theorem 1.1.

A Coverage of the labelled-tree enumeration

Given a labelled tree of order $n \geq 2$, repeatedly delete its least-labelled current leaf and record that leaf's neighbor, stopping when two vertices remain. Leaf deletion preserves connectedness and creates no cycle, so this is defined throughout. It records a sequence of length $n-2$, the Prüfer code; the two surviving vertices are adjacent.

To decode a sequence $a_1, \ldots, a_{n-2}$, initialize

$$d_i = 1 + |\{j : a_j = i\}| \quad (0 \leq i < n).$$

At step j, choose the least label u with current $d_u = 1$, add ua_j, and decrease d_u and d_{a_j} by one. Finally join the two labels with current degree value one. Removed vertices have degree value zero.

When m labels are active, the remaining code has length $m-2$. Each active degree value is one plus the occurrences of its label in that code, and removed labels have no remaining occurrence. This invariant holds initially and is preserved at each step. At least two active labels have no occurrence, because $m-2$ symbols cannot contain all but one of m labels. The next symbol a_j has a remaining occurrence, so it is active with value at least two and differs from the chosen u. Removing u loses no later symbol. Thus the decoder is defined at every step and ends with exactly two active labels of value one.

Reverse the construction from its final edge. Each reverse step attaches a new vertex by one edge to an already present vertex, preserving connectedness and acyclicity. The result is a tree. In the active graph at any stage, each label has one incident edge at its own later removal or at the final pair, and one additional incident edge per occurrence in the remaining code. Its actual degree equals the maintained degree value. Encoding the decoded tree therefore chooses the same least leaf at each step and returns the starting code.

Conversely, encoding a tree records label i once for each incident edge deleted before i is removed, or before the final edge remains. Exactly one incident edge survives to that point. Hence its code contains i exactly $\deg_T(i) - 1$ times. Decoder initialization recovers the actual degrees; the first least leaf and its recorded neighbor agree. After deleting that leaf, the same argument applies to the smaller tree. Induction recovers all edges, including the final edge. The procedures are mutual inverses, so the n^{n-2} sequences give exactly that many labelled trees, with no repetition or omission.

B The exact four-state optimizer

Root a tree at any vertex r. For $a, b \in \{0, 1\}$, let $C_v(a, b)$ be the minimum size of $S \subseteq V(T_v)$ with these conditions:

1. $v \in S$ if and only if $a = 1$;
2. every vertex of T_v other than v has a neighbor in S;
3. v has a selected child if and only if $b = 1$.

Domination of v is not required at this stage; its parent may eventually supply a neighbor. An infeasible state has cost $+\infty$ and is omitted in the implementation. At a leaf the values are $C_v(0, 0) = 0$, $C_v(1, 0) = 1$, and $C_v(0, 1) = C_v(1, 1) = +\infty$.

Let $w_1, \ldots, w_m$ be the children of v and fix a. Choose feasible child states (a_i, b_i). The child w_i is dominated after attaching it to v exactly when

$$b_i \vee a = 1. \tag{10}$$

Its neighbors are its children and v; b_i records a selected child and a the selection of v. Its own selection is irrelevant to its domination. Therefore

$$C_v(a, b) = a + \min\left\{ \sum_{i=1}^{m} C_{w_i}(a_i, b_i) : b_i \vee a = 1 \text{ for every } i,\ \bigvee_{i=1}^{m} a_i = b \right\}. \tag{11}$$

An empty disjunction is zero; a minimum with no feasible choice is $+\infty$.

For the lower-bound direction, restrict any feasible set at v to each child subtree. Every proper descendant of w_i has all its neighbors in that subtree, by uniqueness of paths, and is therefore dominated within the restriction. It defines a feasible child state. Domination of w_i gives (10), and the selected-child condition at v gives the disjunction constraint. Disjointness of child subtrees gives total size a plus the child sizes. Replacing each size by its state minimum proves the lower bound.

For the upper-bound direction, choose minimum sets for a feasible collection of child states and include v exactly when $a = 1$. Proper descendants are dominated by their state conditions, children by (10), and the disjunction gives the selected-child condition at v. The union is feasible with the displayed cost. This proves (11) in both directions.

At the root no parent can dominate it, so a selected child is necessary:

$$\gamma_t(T) = \min\{C_r(0, 1), C_r(1, 1)\}. \tag{12}$$

Every total dominating set restricts to one of these root states, and every feasible root state gives a total dominating set, proving (12).

The implementation processes children one at a time. For fixed a, the initial cost is a and the initial disjunction is zero. A compatible child state adds its cost and changes the running disjunction to its disjunction with a_i. The least cost is retained for each resulting bit. This implements repeated minimization of exactly the same finite expression. Each cost carries the union of its selected-vertex masks; the final mask is checked against (9) and its cardinality against the cost. All calculations are integer calculations without a numerical optimization tolerance.

C Source identification and literature scope

The primary source identifying the target is the 23 July 2026 archival capture of DeLaViña's WOWII page [1], retrieved on 13 October 2026. Entry 348 is dated 18 February 2009 and has historical marker `0`. Its original linked definitions are 94 (total domination), 46 (neighborhood of a vertex subset), and 66 (set difference); the degree-two set is defined in the entry. These identify (1) with its quantifiers and open-neighborhood convention. The source record retains the row, definition excerpts, URLs, retrieval dates, and SHA-256 digests. Two later public transcriptions agree, but both derive from the collection and are not independent mathematical reviews.

The live original curator endpoint was not reached. The archival marker and a current secondary catalogue do not establish the current original curator status. The full text of Desormeaux–Henning [3] was not available in the literature check; only its bibliographic record and abstract were inspected. Its possible relationship to this exact inequality is unchecked. The paper proves the recorded mathematical statement and the displayed strengthening; worldwide first priority is not established by the bounded source search.

Acknowledgment and assistance disclosure

OpenAI Codex assisted in developing the proof, drafting the exposition, and preparing the supplementary programs. It is not listed as an author. The package records self-checks and a reproducible finite replay, rather than external peer review.

Data and code availability

The supplementary archive contains the complete LaTeX source, compiled manuscript, standard-library Python verifier and replay driver, frozen expected results, source metadata, and a SHA-256 manifest. These files suffice to read the proof and run the verification without an internal archive. No public repository or DOI for this manuscript is asserted.

References

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